

Q5		Suggested Solutions	Mark
5	(a)	e.m.f. is the energy per unit charge converted <i>from other forms of energy to electrical energy</i> as charge is driven through a whole circuit.	B1
		Potential difference <i>between two points in a circuit</i> is the energy per unit charge converted <i>from electrical energy to other forms of energy</i> when charge passes between the two points.	B1
	(b)	Since this is a resistance wire of uniform resistivity and cross-sectional area (by the formula $R = \rho l / A$), the resistance R is proportional to the distance l between the two mentioned points.	B1
		Since a constant current flows through the whole circuit, the potential difference between two points on the resistance wire is proportional to resistance R and proportional to the distance l between the two mentioned points.	B1
	(c)	(i) When CK is 60.0 cm, AJ is 54.0 cm. Since ammeter reading is zero, p.d. across CK = p.d. across AJ $= V_{AB} (AJ / AB)$ $= 12.0 (54.0 / 120.0)$ $= 5.4 \text{ V}$	B1 B1
		(ii) When CK = 60.0 cm, resistance across CK = 4.5Ω . When AJ = 54 cm, p.d. across AJ = 5.4 V p.d. across AJ = p.d. across CK $5.4 = \frac{4.5}{4.5 + r} E \dots\dots\dots(1)$ When CK = 20.0 cm, resistance across CK = 1.5Ω . When AJ = 45 cm, p.d. across AJ = 4.5 V p.d. across AJ = p.d. across CK $4.5 = \frac{1.5}{1.5 + r} E \dots\dots\dots(2)$ Eqns [1]/[2], $1.2 = \frac{3(1.5 + r)}{4.5 + r}$ $5.4 + 1.2r = 4.5 + 3r$ $r = 0.5 \Omega$ By eqn [1],	M1 M1 M1

			$5.4 = \frac{4.5}{4.5 + 0.5} E$ $E = 6.0 \text{ V}$	A1
Qn 6	Suggested Solutions			Mark
6(a)				B3